

Gemma 4 Technical Report

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ABSTRACT

We introduce Gemma 4, a new generation of open-weight, natively multimodal language models in the Gemma model family. Designed to advance compute efficiency and reasoning, the Gemma 4 model suite features dense and Mixture-of-Experts architectures, ranging from 2.3B to 31B parameters. Alongside improved vision and audio encoders for all model sizes, we propose a unified, encoder-free architecture for our 12B model, which ingests raw audio and image patches. Furthermore, we integrate a thinking mode, enabling Gemma models to generate reasoning traces prior to responding. We improve inference speed, memory, and compute efficiency, as well as long-context abilities through critical design choices. Gemma 4 establishes a leap in performance across STEM, multimodal, and long-context benchmarks, and rivals larger, frontier open models in human-rated tasks.

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Gemma 4 Technical Report

Gemma Team, Google DeepMind¹

We introduce Gemma 4, a new generation of open-weight, natively multimodal language models in the Gemma model family. Designed to advance compute efficiency and reasoning, the Gemma 4 model suite features dense and Mixture-of-Experts architectures, ranging from 2.3B to 31B parameters. Alongside improved vision and audio encoders for all model sizes, we propose a unified, encoder-free architecture for our 12B model, which ingests raw audio and image patches. Furthermore, we integrate a thinking mode, enabling Gemma models to generate reasoning traces prior to responding. We improve inference speed, memory, and compute efficiency, as well as long-context abilities through critical design choices. Gemma 4 establishes a leap in performance across STEM, multimodal, and long-context benchmarks, and rivals larger, frontier open models in human-rated tasks.

1. Introduction

The rapid evolution of large language models has driven the need for open-weight models with strong multimodal understanding, reasoning, and computational efficiency. Building upon the foundations of its predecessors (Gemma Team, 2024a,b, 2025a), we introduce Gemma 4, the most capable and efficient generation in the Gemma model family to date. Gemma 4 offers natively multimodal architectures, capable of seamlessly processing text, images, and audio while achieving frontier-level performance on highly complex reasoning tasks. The Gemma 4 family is built to serve a variety of on-device hardware. The model suite includes both dense architectures (2.3B, 4.5B, 12B, and 31B parameters) and a Mixture-of-Experts (Jacobs et al., 1991, MoE) variant with 3.8B activated and 26B total parameters. We introduce several architectural and methodological innovations:

- **Thinking mode for advanced reasoning:** We introduce a thinking mode (OpenAI, 2024) to Gemma 4 models. By outputting a reasoning trace before the response, models demonstrate improved capabilities in reasoning-heavy domains such as mathematics and coding.
- **Long-context efficiency:** Extended contexts lead to a memory explosion in the KV cache. We conserve a 5:1 ratio of local sliding window to global self-attention (4:1 for the 2.3B model) and use p -RoPE (Barbero et al., 2025) as positional encoding. Combined with KV cache

sharing (Shazeer, 2019) and the reuse of keys as values in global layers (Kayyam et al., 2026), these optimizations reduce the global KV cache footprint by up to 37.5%.

- **Compute efficiency:** We release an autoregressive multi-token prediction (MTP) drafter head (Li et al., 2024) designed for speculative decoding (Leviathan et al., 2023) to improve the decoding speed of our models.
- **Memory efficiency:** We provide quantized versions of our models trained with quantization-aware training (Jacob et al., 2018, QAT) to reduce their parameter memory footprint and latency with minimal impact on quality.
- **Encoder-free architecture:** Gemma 4 models have frozen vision and audio encoders. We introduce a unified encoder-free architecture for the 12B model, which projects raw 40ms audio chunks and image patches into the LLM embedding space, alleviating the need for separate encoders and reducing memory fragmentation.

In this technical report, we outline the different model architectures across model sizes as well as the pre-training and post-training recipe of Gemma 4. Through comprehensive benchmarks and human evaluations such as Arena (Chiang et al., 2024), we demonstrate that Gemma 4 operates at a level comparable to larger, frontier open-source models across text, image, and audio modalities. We release the Gemma 4 models under an Apache 2.0 license, empowering developers and researchers everywhere to build upon, customize, and extend these capabilities.

¹See Contributions and Acknowledgments section for full author list. Please send correspondence to gemma4report@gmail.com.
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Model	Audio Encoder	Vision Encoder	Embedder	Einsums	Drafter
E2B	305M	150M	400M + 2,340M	1,870M	76M
E4B	305M	150M	670M + 2,820M	3,940M	77M
12B	-	-	1,000M	10,890M	400M
26B-A4B*	-	550M	740M	24,500M / 2,800M (active)	430M
31B	-	550M	1,410M	29,290M	500M

Table 1 | Parameter counts for the Gemma 4 models. The vocabulary we use has 262k entries. The model noted with a star is an MoE defined by its number of active parameters. Note that the extra embedder parameters in E2B and E4B are per-layer embeddings.

2. Model Architecture

Gemma 4 models follow a decoder-only Transformer architecture (Vaswani et al., 2017). Our models have pre-norm and post-norm with RMSNorm (Zhang and Sennrich, 2019), and QKNorm (Henry et al., 2020).

Dense and MoE: The Gemma 4 family of models comprises dense architectures, with effective 2.3B (**E2B**), effective 4.5B (**E4B**), **12B** and **31B** parameters, as well as an MoE model with 3.8B activated parameters for 26B total parameters (**26B-A4B**). E2B and E4B use per-layer embeddings as in Gemma 3n (Gemma Team, 2025b), making them 2.3B and 4.5B effective out of 5B and 8B total parameters respectively.

Model	TPU	#Chips	Shards		
			Data	Seq	Replica
E2B	v6e	4,096	16	8	32
E4B	v6e	6,144	16	16	24
12B	v5p	12,288	16	16	48
26B-A4B*	v6e	6,144	16	16	24
31B	v6e	10,240	16	16	40

Table 2 | Pre-training infrastructure with sharding by data, sequence (Seq), and replica.

Long-context efficiency: Our local to global attention ratio patterns follow Gemma Team (2025a), that is, 4-to-1 local attention blocks for E2B and 5-to-1 for the rest. We improve memory efficiency by re-using keys as values in the global attention layers (except in E2B and E4B), *i.e.*, values = keys. We encode position

with p -RoPE with $p = 0.25$ on global attention layers and with RoPE on local attention layers, effectively reducing the global KV cache by 37.5%. The RoPE frequencies are set to 1M and 10k on global and local attention layers, respectively. Finally, we share the KV cache with ratios of 20/35 and 18/42 for the E2B and E4B model.

2.1. Vision modality

E2B and E4B Gemma models come with a 150M vision encoder, while larger models use a 550M encoder (except for the unified 12B). Both are Vision Transformers (Dosovitskiy et al., 2021, ViT) with a patch size of 16, whose architectural differences are detailed in Table 10 in Appendix. Our vision encoders support variable aspect ratios (see Figure 2 and Algorithm 1) and incorporate both axial 2D-RoPE (Heo et al., 2024) with non-causal attention and 2D absolute positional embeddings. We restrict the maximum number of tokens, $N_{\max}$ to the values 70, 140, 280, 560 and 1120 (see Algorithm 1 for implementation details).

2.2. Audio modality

E2B and E4B Gemma models use a 305M audio encoder that processes audio in 40ms chunks with Mel filterbank inputs. The encoder architecture is based on the Universal Speech Model (Zhang et al., 2023, USM), consisting of two downsampling convolution layers followed by twelve Conformer layers (Gulati et al., 2020). While the architecture remains similar to that of Gemma 3n, we reduce the number of parameters by 55% (from 680M to 305M). We do not use vector quantization; the LLM ingests the con-

tinuous representations produced by the audio encoder. As with the vision encoder, we keep weights frozen during pre-training.

2.3. Encoder-free architecture

Gemma 4 12B is trained from scratch based on a new, unified, and encoder-free model paradigm, replacing the separate vision and audio encoders with lightweight projection modules. For the vision modality, Gemma 4 12B takes in $48 \times 48 \times 3$ RGB patches, but replaces the 550M vision encoder by a single large matmul (35M parameters). Spatial awareness is maintained by adding 2D coordinate-based positional embeddings directly to the patch representations before a final LayerNorm layer (Ba et al., 2016).

For audio, the 305M USM-based conformer encoder is *entirely discarded*. Raw audio is segmented into 40ms chunks at 16kHz, resulting in 640-dimensional vectors per chunk. These are projected directly into the LLM embedding space. Since audio is a temporal sequence, it does not require additional positional encoding.

Model	bf16	Quantized	KV Cache
E2B	4.6	0.8 [†]	+0.05
E4B	9.0	2.3 [†]	+0.14
12B	24.0	7.65 [‡]	+0.28
26B-A4B*	52.0 / 7.6	16.2/2.8 [‡]	+0.28
31B	64.0	19.2 [‡]	+1.10

Table 3 | Text only, Gb memory footprint comparison between raw and quantized checkpoints for weights and int8 KV caching (+KV) at 32k context size. [†] is mobile quantization, [‡] is Q4_0.

2.4. Pre-training

We follow a similar pre-training as Gemma 3.

Training data. Our pre-training dataset is a large-scale, diverse collection of data from a wide range of domains and modalities, including web documents, code, images, and audio (for E2B, E4B and 12B), with a cutoff date of January 2025.

Tokenizer. We use the same tokenizer as Gemma Team (2025) that is, a SentencePiece tok-

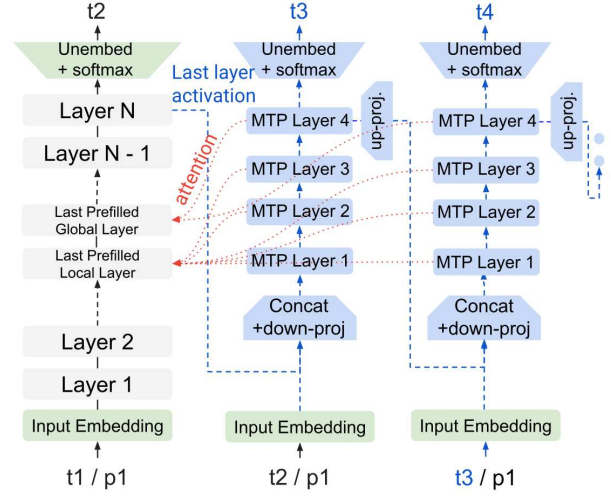


Figure 1 | The autoregressive MTP drafter (blue blocks on the right) is fed activations and KV cache from the main model (grey blocks).

enizer (Kudo and Richardson, 2018) with split digits, preserved whitespace, and byte-level encodings. The vocabulary has 262k entries.

Filtering. We filter data to decontaminate benchmarks, and to reduce the risk of unwanted or unsafe utterances and the risk of recitation.

2.5. Quantization-Aware Training

We provide quantized models and encoders in different formats along with the raw checkpoints. Based on the most popular open source quantization inference engines (e.g. llama.cpp) as well as efficient hardware support, we focus on two sets of weight representations:

- mobile quantization: per-channel low bitwidth weight (mix of int2 and int4) and activation quantization (int8).
- Q4_0 quantization: blockwise quantization, often referred to as Q4_0.

In Table 3, we report the memory filled by raw and quantized models with and without a KV cache for a sequence of 32k tokens. Furthermore, to enable stable inference in fp16, we introduce a scalar scale at each block in order to bound the activation ranges to fit fp16.

Rank	Model	Elo	95% CI	Open	Type	#params/#activated
1	Claude Fable 5	1508	± 9	no	-	- / -
...						
15	GLM 5.1	1475	± 6	yes	MoE	744B / 40B
25	GLM 5.2 (Max)	1471	± 10	yes	MoE	744B / 40B
29	MiMo V2.5 Pro	1466	± 5	yes	MoE	1T / 42B
34	Kimi K2.6	1460	± 5	yes	MoE	1T / 32B
36	DeepSeek V4 Pro Thinking	1458	± 5	yes	MoE	1.6T / 49B
37	GLM 5	1457	± 5	yes	MoE	744B / 40B
38	DeepSeek V4 Pro	1456	± 5	yes	MoE	1.6T / 49B
43	Gemma 4 31B	1451	± 8	yes	Dense	31B
44	Kimi K2.5 Thinking	1450	± 4	yes	MoE	1T / 32B
57	Qwen 3.5 397B-A17B	1444	± 4	yes	MoE	397B / 17B
61	Gemma 4 26B-A4B	1438	± 8	yes	MoE	26B / 4B
63	DeepSeek V4 Flash Thinking	1436	± 5	yes	MoE	284B / 13B
...						
157	Gemma 3 27B	1366	± 4	yes	Dense	27B

Table 4 | Leading open-weight models on Arena Text (Chiang et al., 2024) (as of June 19, 2026). Models are evaluated through blind side-by-side evaluations by human raters, and attributed scores based on the Elo rating system. The top closed model (gray) is included for scale. Gemma models rival much larger models, and Gemma 4 31B is the leading dense open model on the leaderboard.

We also apply QAT to the image and audio encoders. On the 150M image encoder, quantizing activations and weights to 8-bit precision (W8A8) yields a $2\times$ reduction in total forward-pass memory footprint (from 400 MB to 200 MB, including on-device compilation overhead) and a 44% reduction in on-device latency relative to Gemma 3n on newer hardware. On the audio encoder, we further reduce activation precision to 8 bits and weight precision to $\{2, 4, 8\}$ bits, varying by layer cluster. Overall, we achieve a 78% reduction in on-disk footprint, from 390 MB in Gemma 3n to 87 MB in this version.

2.6. Multi-Token Prediction Drafter

We train a small autoregressive MTP drafter head with our models, used for speculative decoding. In our MTP procedure, the model’s last layer activations from the previous step and token embeddings are fed into the MTP head. The MTP head generates future tokens sequentially using a separate embedder and a 4-layer Transformer block that cross-attends to the KVs of the main model (Figure 1), thus eliminating the need for

MTP prefill and supporting any draft length. The Transformer block has model dimension 256 for E2B and E4B, 1024 for 26B-A4B and 31B, three local, and one global attention layers.

Efficient MTP Decoding. For the E2B and E4B drafters, we reduce the decoding overhead by replacing the projection operation to the entire vocabulary by a top-k operation on clusters of tokens. As a result, final matrix multiplication is reduced from $d \times 262,000$ to $d \times 4096$ while preserving a similar acceptance rate.

2.7. Compute Infrastructure

We train our models with TPUv5p and TPUv6e as outlined in Table 2. Each model configuration is optimized to minimize training step time. For our larger models, we leverage Slice-Granularity Elasticity (Gemini Team, 2025), which allows continuous training with fewer “slices” of TPU chips when there is a localized failure. This reconfiguration reduces the delay caused by interruptions from many minutes to a few seconds.

	Gemma 4					Gemma 3
	31B	26B-A4B	12B	E4B	E2B	27B non-thinking
MMLU Pro	85.2	82.6	77.2	69.4	60.0	67.6
AIME 2026 no tools	89.2	88.3	77.5	42.5	37.5	20.8
LiveCodeBench v6	80.0	77.1	72.0	52.0	44.0	29.1
Codeforces Elo	2150	1718	1659	940	633	110
SciCode	43.0	40.0	38.0	24.0	21.0	21.0
GPQA Diamond	84.3	82.3	78.8	58.6	43.4	42.4
Big Bench Extra Hard micro avg	74.4	64.8	53.0	33.1	21.9	19.3
HLE	19.5	8.7	5.2	-	-	-
HLE with search	26.5	17.2	-	-	-	-
IFBench	76.0	72.0	74.0	44.0	38.0	32.0
IFEval	98.9	98.5	97.2	96.7	94.6	90.4
MMMLU	88.4	86.3	83.4	76.6	67.4	70.7
MRCR v2 8-needle, 128k	66.4	44.1	43.4	25.4	19.1	13.5
Terminal Bench Hard	36.0	14.0	18.0	8.0	3.0	4.0
Tau2 – airline	75.0	76.0	75.0	52.0	31.0	39.0
Tau2 – retail	86.4	85.5	77.6	67.1	34.6	6.6
Tau2 – telecom	69.3	43.0	54.4	18.4	19.7	3.1

Table 5 | Performance comparison of Gemma 3 27B and Gemma 4 models on diverse benchmarks. All models are in thinking mode unless explicitly stated.

The optimizer state is sharded using an implementation of ZeRO-3 (Ren et al., 2021). For multi-pod training, we perform a data replica reduction over the data center network, using the Pathways approach of Barham et al. (2022). We use the single controller programming paradigm of JAX (Roberts et al., 2023) and Pathways, along with the GSPMD partitioner (Xu et al., 2021) and the MegaScale XLA compiler (XLA, 2019).

3. Instruction Tuning

Pre-trained models are turned into instruction-tuned models with a similar post-training approach as in Gemma 3. A significant difference is the addition of a thinking mode, where the model can output a reasoning trace before answering.

Data filtering. We carefully optimize the data used in post-training to maximize model perfor-

mance. We filter examples that show certain personal information, unsafe or toxic model outputs, mistaken self-identification data, and duplicated examples. Including subsets of data that encourage better in-context attribution, hedging, and refusals to minimize hallucinations also improves performance on factuality metrics, without degrading model performance on other metrics.

PT versus IT formatting. All models share the same tokenizer, with some control tokens dedicated to IT formatting. A key difference is that PT models output an `<eos>` token at the end of generation, while IT models output `<turn|>` at the end of the generation. An example is given for IT in Table 11. Fine-tuning either model type thus requires adding their respective end tokens. We detail how to activate thinking and how models handle function calling in Table 11.

	Gemma 4					Gemma 3
	31B	26B-A4B	12B	E4B	E2B	27B
MMMU Pro	76.9	73.8	69.1	52.6	44.2	49.7
MATH-Vision	85.6	82.4	79.7	59.5	52.4	46.0
MedXPertQA MM	61.3	58.1	48.7	28.7	23.5	-
InfographicVQA	92.0	89.3	88.4	70.0	63.9	70.6
OmniDocBench 1.5 ↓	0.131	0.149	0.164	0.181	0.290	0.365

Table 6 | Gemma 4 models performance on vision benchmarks at different resolutions (thinking). We use the maximal supported resolution (1120 vision tokens) and report results with 280 vision tokens in Table 12. Gemma 3 27B is non-thinking and uses Pan & Scan.

4. Evaluation of final models

In this section, we evaluate the IT models over a series of automated benchmarks and human evaluations across a variety of domains, as well as static benchmarks such as MMLU Pro.

4.1. Human evaluation

We report the performance of our 31B and 26B-A4B models on Arena (Chiang et al., 2024) in blind side-by-side evaluations by human raters against other state-of-the-art models. We report Elo scores in Table 4. Gemma 4 31B is the top open model in the dense category, and both Gemma 4 31B and 26B-A4B show performance equal to much larger open models.

4.2. Static benchmarks

In Table 5, we show the performance of our final models across a variety of benchmarks compared to Gemma 3 27B. Gemma 4 31B is closest in size and significantly better across the board, while E2B roughly matches Gemma 3 27B performance with 10x less parameters. Table 6 shows the performance of Gemma 4 models on vision benchmarks, with E4B equaling or outperforming Gemma 3 27B on all evals. Tables 7 and 8 display the multilingual audio transcription and translation performance of E2B & E4B and of 12B respectively. Table 9 shows a leap on long-context capabilities between Gemma 3 27B and Gemma 4 models, with E4B outperforming Gemma 3 27B.

5. Responsibility, Safety, Security

As open models become central to enterprise infrastructure, provenance and security are paramount. Gemma 4 undergoes the same rigorous safety evaluations as Gemini models. Responsibility, safety, and security are of utmost importance in the development workflow, ensuring that these language models are designed from the ground up for responsible AI development.

5.1. Governance & Assessment

Our approach to assessing the benefits and risks of Gemma 4 reflects the foundation established in prior models, updated to account for its expanded multimodal capabilities. We maintain the belief that openness in AI can spread the benefits of these technologies across society, but this must be continuously evaluated against the risk of malicious uses that can cause individual and institutional harm (Weidinger et al., 2021).

Gemma 4 models were developed in partnership with internal safety and responsible AI teams. Releasing these models required careful scrutiny of the evolving risks associated with LLMs and an understanding of how models are deployed in the wild. While an open model shares innovation across the AI ecosystem, we remain committed to providing educational resources to users and monitoring downstream model usage.

<i>CoVoST (CorpusBLEU ↑)</i>										
	Params	Size	ja → en	de → en	fr → en	es → en	it → en	ru → en	zh → en	AVG
Gemma 4 E2B	305M	87 MB	21.4	39.2	39.2	43.2	40.8	46.4	17.9	35.4
Gemma 4 E4B			25.5	42.0	41.0	44.8	43.0	49.4	21.9	38.2
Gemma 3n E2B	680M	390 MB	17.7	36.5	35.7	39.9	38.5	39.2	13.9	31.6
Gemma 3n E4B			22.3	39.1	38.4	41.8	40.4	43.7	17.4	34.7

<i>FLEURS ASR (WER ↓, * = CER ↓)</i>													
	en	ko*	ja*	de	fr	hi	es	it	pt-br	ru	ar	zh*	AVG
Gemma 4 E2B	0.080	0.066	0.107	0.076	0.101	0.101	0.042	0.041	0.056	0.084	0.143	0.187	0.090
Gemma 4 E4B	0.065	0.053	0.078	0.061	0.080	0.086	0.035	0.032	0.046	0.068	0.162	0.136	0.075
Gemma 3n E2B	0.076	0.101	0.163	0.079	0.130	0.106	0.051	0.044	0.067	0.112	0.131	0.235	0.108
Gemma 3n E4B	0.066	0.073	0.111	0.065	0.098	0.089	0.041	0.034	0.053	0.087	0.101	0.203	0.085

Table 7 | Audio performance for Gemma 4 and Gemma 3n models. Top: CoVoST (S2TT prompt: *transcribe then translate*). Bottom: FLEURS ASR (transcription). Compared to Gemma 3n of corresponding sizes, Gemma 4 achieves a 12% (E2B) / 10% (E4B) relative improvement on translation and a 17% (E2B) / 12% (E4B) relative improvement on transcription, despite a 78% reduction in on-disk audio encoder footprint (from 390 MB to 87 MB after quantization).

<i>FLEURS ASR (WER ↓, * = CER ↓)</i>				
en	ko*	ja*	de	fr
0.063	0.057	0.080	0.053	0.081
es	it	pt-br	ru	ar
0.038	0.030	0.047	0.068	0.070

<i>CoVoST (XX → EN, CorpusBLEU ↑)</i>					
ja	de	fr	es	it	ru
26.4	41.9	42.5	44.6	43.3	50.5

Table 8 | Audio performance of Gemma 4 12B model on supported languages, demonstrating that competitive audio-text performance can be achieved without a dedicated audio encoder.

5.2. Safety Policies and Train-Time Mitigations

A key pillar of Gemma’s safety approach is aligning our fine-tuned models with Google’s AI principles and safety policies. These policies aim to prevent our generative models from producing harmful content, specifically:

- Content related to child sexual abuse material

(CSAM) and exploitation;

- Dangerous content, e.g., promoting suicide, or instructing in activities that could cause real-world harm;
- Sexually explicit content;
- Hate speech, e.g., dehumanizing members of protected groups;
- Harassment, e.g., encouraging violence against people.

To mitigate these risks, Gemma 4 models underwent careful input data pre-processing and scrutiny. The training data was specifically filtered for the removal of certain personal information and other sensitive data to guard against privacy violations. Post-training evaluations and train-time mitigations were also implemented to align the model with our safety policies.

5.3. Safety Evaluations

We conduct rigorous automated and human evaluations to understand the potential harms our models might cause. For all areas of safety testing, we saw major improvements in every category of content safety relative to previous Gemma models. Overall, Gemma 4 models significantly out-

Benchmark	Metric	Context length	Gemma 4					Gemma 3
			31B	26B-A4B	12B	E4B	E2B	27B
RULER	Accuracy	32k	96.8	97.3	96.4	95.2	83.0	91.1
		128k	96.4	89.8	91.2	86.6	70.4	66.0
LOFT Text Retrieval	Recall@k	128k	79.5	66.3	66.4	58.5	50.5	8.6
GraphWalks	F1	<128k	82.3	72.6	71.0	50.9	4.1	32.8
MTOB (eng→kgv)	chrF	~128k (Half book)	52.9	50.0	45.1	37.8	15.4	41.0
		~256k (Full book)	54.3	48.9	41.9	-	-	-
MTOB (kgv→eng)	chrF	~128k (Half book)	48.6	45.0	37.3	34.6	28.2	31.2
		~256k (Full book)	46.2	42.7	32.9	-	-	-

Table 9 | Long context performance of Gemma 3 and Gemma 4 models (without thinking).

perform Gemma 3 and 3n models in improving safety, while keeping unjustified refusals low.

Importantly, all testing was conducted without safety filters to accurately evaluate the model’s inherent capabilities and behaviors. For both text-to-text and image-to-text modalities, and across all model sizes, the models produced minimal policy violations. We balance development speed with targeted safety testing, upholding the commitments laid out in our Frontier Safety Framework (Google DeepMind, 2024).

5.4. Ethical Considerations and Risk Mitigation

The development of LLMs introduces specific ethical considerations. In making Gemma 4, we focused heavily on:

- **Bias and Fairness:** LLMs trained on large-scale text and image data can reflect embedded socio-cultural biases. We encourage developers to perform continuous monitoring (using evaluation metrics and human review) and explore debiasing techniques during model fine-tuning.
- **Misinformation and Misuse:** LLMs can be misused to generate false or misleading text. We provide technical limitations, developer education, and guidelines for responsible use within the Responsible Generative AI Toolkit to mitigate malicious applications.

- **Privacy Considerations:** While our training datasets were filtered to remove certain personal information and other sensitive data, developers are strongly encouraged to adhere to local privacy regulations and implement privacy-preserving techniques in their applications.

5.5. Our Approach to Responsible Open Models

Designing safe, secure, and responsible applications requires a system-level approach that mitigates risks associated with specific use cases and environments. We provide guidelines, mechanisms, and safeguards for content safety, and encourage developers to implement appropriate configurations based on their product policies. We will continue to adopt safety mitigations proportionate to potential risks, sharing these models with the community only when confident that the benefits significantly outweigh foreseeable risks.

6. Discussion and Conclusion

In this technical report, we presented Gemma 4, an open-weight model family featuring multimodal dense and MoE architectures designed for varied hardware environments. Gemma 4 models come with a thinking mode in which they generate reasoning traces prior to responding, improving overall performance. We introduced a unified, encoder-free architecture that processes

raw audio and image patches. We also alleviated long-context memory limitations via better local-to-global attention ratios, positional encoding, and KV cache sharing. We increased the overall compute efficiency via QAT and memory efficiency via MTP drafters. Gemma 4 models demonstrate a leap in performance compared to Gemma 3 across benchmarks, and human evaluations demonstrate that Gemma 4 performs comparably to significantly larger open models, providing a scalable foundation for edge deployment and reasoning while supporting open research.

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Appendix

Conversation format. We give an example of a conversation including thinking, function definition and function calling in Table 11.

Vision. We detail the vision encoder architecture in Table 10. We then illustrate how images are resized before being fed to the vision encoder in Figure 2, and detail the resizing algorithm in Algorithm 1. We display the vision benchmark scores of Gemma 4 models at low resolution ($N_{max} = 280$) in Table 12.

Total Params	d_{model}	d_{MLP}	N_{heads}	N_{layers}
550M	1152	4304	16	27
150M	768	3072	12	16

Table 10 | Vision encoder architecture.

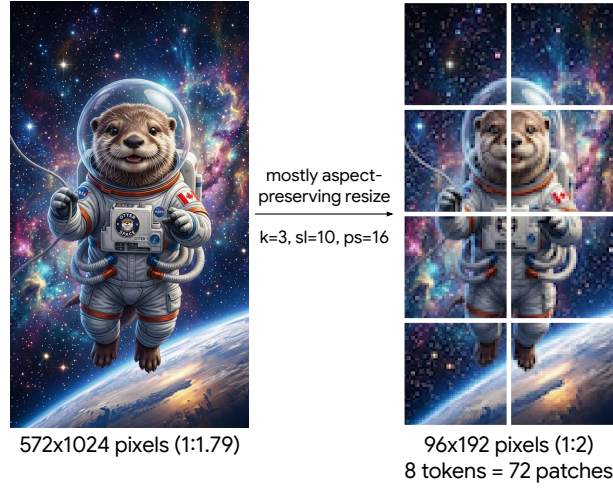


Figure 2 | Image resizing. Here we use `patch_size=16`, `pooling_kernel_size=3`, `max_soft_tokens=10`. The image is thus first resized to 2×4 pooled patches (each of size 48px^2), which is the closest match that results in a sequence length below the targeted 10. The 72 patches (each of size 16px^2) are then processed by the vision encoder, the vision encoder representations are pooled 3×3 , and the resulting 8 soft tokens are processed by the LLM backbone.

Algorithm 1 Aspect-Ratio Preserving Image Resizing (see also Figure 2)

Require: Image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, patch size p , max tokens N_{max} , pooling kernel size k

- 1: $m \leftarrow k \cdot p$ ▷ Pooled patch size
 - 2: $T \leftarrow N_{max} \cdot m^2$
 - 3: $f \leftarrow \sqrt{T/(H \cdot W)}$ ▷ Ideal scaling factor
 - 4: $H_{ideal} \leftarrow f \cdot H$
 - 5: $W_{ideal} \leftarrow f \cdot W$
 - 6: $H_{target} \leftarrow \lfloor H_{ideal}/m \rfloor \cdot m$ ▷ Round down
 - 7: $W_{target} \leftarrow \lfloor W_{ideal}/m \rfloor \cdot m$
 - 8: $\mathbf{I}_{resized} \leftarrow \text{BicubicResize}(\mathbf{I}, H_{target}, W_{target})$
 - 9: **return** $\mathbf{I}_{resized}$
-

Context	Formatting
Thinking toggle	<code>< think ></code>
Function declaration	<code>< tool>declaration:...<tool ></code>
Function call	<code>< tool_call>call:...<tool_call ></code>
Thinking trace	<code>< channel>thought ...<channel ></code>
System turn	<code>< turn>system</code>
User turn	<code>< turn>user</code>
Model turn	<code>< turn>model</code>
End of turn	<code><turn ></code>
Example of discussion:	
Toggle thinking mode. Declare function. User: I want you to book a train ticket for me. Model: <...> Where would you like to go? User: To Rome. Model: <...> Looking for available tickets: <function call>	
Model input:	
<pre> [BOS] < turn>system < think > < tool>declaration:search_train{...}<tool ><turn > < turn>user I want you to book a train ticket for me.<turn > < turn>model < channel>thought ...<channel >Where would you like to go?<turn > < turn>user To Rome.<turn > < turn>model </pre>	
Model output:	
<pre> < channel>thought ...<channel >Looking for available tickets: < tool_call>call:search_train{from:< >Athens< >,to:< >Rome< >}<tool_call ><turn > </pre>	

Table 11 | Formatting for Gemma IT models. Explicitly add the `[BOS]` token after tokenization, or use the `add_bos=True` option in the tokenizer. *Do not tokenize the text "[BOS]"*. Add `<|think|>` in a leading system turn to activate the thinking mode. Check the official documentation for the function declaration and function calling syntax, as well as more advanced examples.

	Gemma 4				
	31B	26B-A4B	12B	E4B	E2B
MMMU Pro	75.8	73.2	67.7	51.4	43.2
MATH-Vision	83.4	80.3	76.7	59.2	53.0
MedXPertQA MM	60.7	55.7	47.4	28.7	22.5
InfographicVQA	82.8	77.8	58.7	54.8	44.6
OmniDocBench 1.5 ↓	0.201	0.269	0.408	0.307	0.496

Table 12 | Gemma 4 models performance on vision benchmarks at resolution $N_{max} = 280$ (thinking).